



Ashok Modhave



Auckland, New Zealand

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NZ Permanent resident

Summary

Ashok is a Senior Full-Stack Software Engineer with 13+ years of experience designing, building, and delivering scalable web and mobile applications across Banking, FinTech, and Travel domains.

He specializes in frontend-heavy full-stack development, with strong expertise in React, React-native, Next.js, Typescript, and modern UI architectures, combined with backend development using Java Spring Boot, Node.js, NestJS, and RESTful microservices. His recent experience focuses on building enterprise-grade web platforms, payment integrations, document generation solutions, Generative AI/LLM powered applications, cloud-native services microservices on AWS and Azure, and monorepo-based frontend ecosystems using NX.

He has a proven track record of owning features end-to-end: from requirements analysis and solution design through implementation, testing, and production deployment, involving functional complexity, time constraints and innovation. He has developed end to end customer facing web applications. His expertise includes developing Restful microservices using Java Spring boot, Node.js, express JS and NestJS and leveraging AWS cloud functionalities like Lambda, DynamoDB, Serverless, S3, SES, EC2, and SNS. He has worked on developing test automation JavaScript framework using playwright, selenium web driver, mocha, chai, assert libraries. He has experience of writing windows and bash scripts and scheduling jobs. He is self-learner and does explore modern technologies in Generative AI, Artificial Intelligence (AI)/Machine Learning (ML) and LLM's.

Recently, he has gained hands-on experience with Generative AI/LLM -based applications, payment processing, and PCI-aware integrations

Ashok is proactive problem-solver, strong collaborator, and mentor who enjoys learning contemporary trends and technologies in web or mobile applications and working on complex systems and delivering high-quality user experiences. He aspires to build a career in New Zealand, leveraging his full stack experience. He is open to exploring new contract opportunities in New Zealand where he can utilize his expertise in skills and grow along with the organization.

Specializing In:

Software Development	Full-stack web application development, frontend architecture, backend API design, microservices, monorepo architecture, technical solution for scalability security-focused development, environment set up, product innovation.
Solution Analysis	Requirement gathering, & technical feasibility, System design, impact analysis, identifying gaps and impacts, estimation, prioritization and architectural decision-making.
Documentation & Testing	RTM, HLD/DLD, API Documentation, coding guidelines and best practices implementation, unit testing, automation testing, TDD-friendly development.
Behavioral	Building teams and executing projects in fast changing technology advancements, Understanding of Team dynamics, mentoring, cross-team collaboration, communication, adaptability, ownership mindset.

Skills

Frontend (Web/Mobile) - React.js, Next.js, React Native - TypeScript, JavaScript - Redux, Redux Toolkit, Jotai, Zustand, React Query, RxJs - React Hook Form, React Final Form, Zod, Storybook - HTML5, CSS3, Sass - Material UI, Tailwind CSS, Radix UI, Mantine UI, Hero UI, Bootstrap	Backend - Java, Spring MVC, Spring Boot - Node.js, NestJS, Express.js - Python, Django REST - C#/.Net core, C++ Databases - MongoDB, mLab, DynamoDB - PostgreSQL, MySQL, Oracle, MS SQL - Kafka, H2, RabbitMQ	Cloud & DevOps - AWS: Lambda, S3, SNS, SES, EC2, EKS, VPC, ELB, IAM, Serverless - Amazon Connect - Azure DevOps - Docker, Kubernetes - CI/CD pipelines
Architecture & Platform - NX Monorepo - Microservices/Micro-frontend - REST/SOAP APIs - System Design - Design patterns, OOPS - Data structure & Algorithms	Monitoring & Observability - CloudWatch - Kubernetes logs - Dynatrace - Sentry - New Relic, Sumo Logic	Testing - React Testing Library, Jest - Selenium WebDriver, playwright, Cypress - Postman, Rest assured
AI & Machine Learning - Generative AI - Azure OpenAI - LLM concepts Agile & Documentation - Azure Boards - JIRA, Confluence	Web servers - Node.js - Tomcat, IIS - Vercel, Heroku	Tools & Others - VS Code, Xcode, Android Studio - Eclipse, IntelliJ - Git, SVN - Postman - Windows & bash scripting, Perl - Google APIs

Employment History

Bank Of New Zealand

Apr 2025 – Present

Company information – Bank of New Zealand, known as BNZ, is a leading bank in New Zealand provides retail, business, and institutional banking services through a nationwide branch and digital network.

Role - Senior Full Stack Developer

Doc-Gen UI, Home Lending Adapter - Web app + Spring boot API

Auckland - NZ Apr 2025 – Present

Project: Web application for document generation platform supporting home lending, along with providing capability for search/retrieve, save/resume, download and persisting onboarding customer data.

Responsibilities

- Leading frontend architecture and development and optimizing the technical depths of the application, improving performance and operational efficiency, currently catering ~70% of document processing volume by replacing the legacy HotDocs platform.
- Building scalable, maintainable React + Typescript application with foundation setup, state management, complex forms, validation and API integration, persistence capability
- Designing and developing Java Spring boot microservices to integrate to internal systems and consume third party providers API like Smart COMM, nCino.
- Partnered with product and UX to translate business requirements into well-structured technical solution.
- Driving code quality via reviews, standards, and continuous improvement.
- Collaborated, mentored and contributed to internal Dev community with solutions and knowledge sharing.

Technology

- React.js, Jotai, React-hook-form, Zod, React-query, Axios

- Java Spring Boot, OpenShift, Kong API, AWS EKS
- Monitoring tools like CloudWatch, Kubernetes logs, Sentry, Dynatrace
- Jenkins, Spinnaker

Serko Ltd.

Oct 2024 – Mar 2025

Company information – Serko Limited, known as Serko, is a travel management and expense technology company, offering connected and frictionless travel experience by innovative, integrated travel and expense solutions.

Role - Senior Front-End Engineer

BBX (Booking for business experience) – Phase 2 - Checkout - Web app **Auckland - NZ** May 2022 – till date

Project: React web application providing solution for end-to-end managed business booking of accommodation for business travels. Phase 2 of this project involved implementing the checkout experience along with payment processing and handling the checkout summary for the property customer is booking.

Responsibilities

- Led frontend development enhancing application for accommodation booking and checkout experience
- Built re-usable component with NX monorepo architecture using Next.js and React.
- Designed and implemented PCI-compliant payment processing flows, integrating with external payment providers using secure iframe-based solutions.
- Review the UX and map it to technical requirements.
- Partnered with backend and platform teams to design scalable API integration patterns.
- Implemented experiment-driven development using Optimizely to A/B test checkout experiences and optimize conversion.
- Improved frontend reliability and observability using New Relic and Sumo Logic.

Technology

- React.js, Next.js, NX Console mono-repo framework
- React query, axios, Orval open API spec generator.
- Monitoring tools like new relic, sumo logic, azure beanstalk

Air New Zealand Ltd.

May 2023 – Oct 2024

Company information – Air New Zealand Limited, known as flag carrier for New Zealand, is an airline company, offering cargo services to, from and within New Zealand.

Role - Senior Software Developer

Secure pay - Web app

Auckland - NZ Feb 2024 – Oct 2024

Project: Next.js web application following NX mono-repo framework providing solution for contact center agents to process the payment over the phone. Customer calling to customer service and want to book some service, so this application provides a platform to allow enter credit card information on your phone and initiate transaction through phone call.

Responsibilities

- Design, review UX, map to technical requirements, implementation and enhancement of application
- Setup the foundation and environment for application
- Integrated frontend with Amazon Connect for call handling workflows.
- Implemented PCI-compliant payment flows by integrating with Sequence Shift payment validator and tokenization services to process payment.
- Implemented Azure B2C authentication for secure authentication and authorization.
- Develop the UI and backend integration.
- Fixed critical call-handling defect preventing agents from ending or restarting calls.
- Improved customer service call-handling efficiency by 43%.

Technology

- React.js, Next.js, react-msal, NX Console mono-repo
- Amazon connects

- React query, react-hook-form, react-msal. Redux toolkit, RxJs
- Monitoring tools like Kubernetes logs, sentry

Tui chatbot - Web app

Auckland - NZ Feb 2024 – Oct 2024

Project: Built Generative AI powered chatbot using Next.js within NX monorepo architecture, providing solution for contact center. This chatbot is capable to train and reads LLM built through internal legal documents and provides the answers to questions of agents.

Responsibilities

- Design, Development and enhance of frontend UI development and backend integration with REST APIs
- Setup the foundation and environment for application
- Implemented the message data streaming and parsing capability using observability with RxJs
- Integrated with Azure OpenAI for LLM-based backend processing and training.
- Designed solution as a universal, domain-agnostic platform where different teams can attach their own LLM datasets.
- Implemented capabilities for suggestion generation, prompt refinement, and LLM training
- Improved agent efficiency by 73% and significantly reduced time to provide correct customer answers.

Technology

- ReactJS, Next.js, NX mono-repo, Generative AI, Azure OpenAI
- React query, react-hook-form, react-msal, RxJs
- Monitoring tools like Kubernetes logs, sentry

Customer feedback form, refund form, pet form, anonymous form - Web app

Auckland - NZ May 2023 – Feb 2024

Project: Customer facing web platform developed using Next.js within NX monorepo framework, customers to submit feedback, refund request, complaints, pet request, anonymous submissions and many more features related to Airpoints.

Responsibilities

- Design, Review UX, map it to technical requirement, develop and enhance project
- Setup the foundation and environment for application
- Develop the UI and integrated REST API with Microsoft Dynamics 365 to automatically create tickets and route them to correct department queues.
- Improved customer service providing self-service digital channels which increased request resolution efficiency and reduced handling time through automation.

Technology

- ReactJS, Next.js, NX Console mono-repo, React-query, react-final-form, Redux toolkit
- Monitoring tools like Kubernetes logs, sentry
- AWS S3, Microsoft dynamics 365

Serko Ltd.

May 2022 – May 2023

Company information – Serko Limited, known as Serko, is a travel management and expense technology company, offering connected and frictionless travel experience by innovative, integrated travel and expense solutions.

Role - Senior Software Developer

Zeno and BBX (Booking for business experience) - Web app

Auckland - NZ May 2022 – May 2023

Project: Next.js NX monorepo web application providing solution for end-to-end managed and unmanaged business booking of flights, car rentals, and hotels.

Responsibilities

- Design, UX review, technical architecture and core contributor to development of project, along with setting up foundation and platform environment
- Designed and established NX monorepo architecture with scalable frontend development using Next.js.

- Built complex booking flows, search results, and itinerary experiences, google Map API integration, component design using Material UI based design system.
- Designed frontend integration strategy using React Query and OpenAPI (Orval) for strongly typed API clients.
- Collaborated with product, UX and backend teams to translate business requirement into technical solution.
- Mentored junior engineers and participated in code reviews and architectural discussions.

Technology

- ReactJS, Next.js, NX Console monorepo, Google Map API
- React query, axios, Orval open API spec generator.
- Monitoring tools like new relic, sumo logic, azure beanstalk

[Datacom Systems Limited](#)

May 2021 – May 2022

Company information – Datacom Group Limited is an Information Technology services company, offering management and consulting, cloud services, ITO, data center services, custom software development, and payroll services.

Role - Senior React Developer

[ERoadDay app](#) - Mobile app

Auckland - NZ June 2021 – May 2022

Project: React native mobile apps for iOS and android to manage the logistic experience of the dashcam placed inside the vehicle, to be able to manage the driving experience and see the statistics

Responsibilities

- Requirement analysis, plan, design, and development of project
- Setup the foundation and environment for application
- Review the UX and map it to technical requirement
- Develop the UI and backend integration

Technology

- React-native, React-Redux, redux toolkit, Testing library

[PGDB \(Plumbing, Gasfitters and Drycleaners board\)](#)- Mobile app

Auckland - NZ May 2021 - June 2021

Project: React-native mobile app responsible for managing the certifications and licenses of the plumbers, Gasfitters and Drycleaners and allowing the renewal and notifying any important follow ups.

Responsibilities

- Bug fixes and support existing app
- Implementation the notification

Technology

- React-native, React-Redux, Testing library

[ASB Bank](#)

Dec 2018 – May 2021

Company information – ASB Bank, a leading bank in New Zealand, was born in this dynamic world of digital, providing its customers with outstanding service and innovative financial solutions like KiwiSaver, investment funds.

Role - Senior Software Engineer

[My KiwiSaver, Investment Fund Withdrawal](#) - Financial investment and saving platform

Auckland, NZ April 2019 - May 2021

Project: New Digital investment platform for retirement or saving for your first home, platforms helps to manage, grow and maximize your savings with the ASB KiwiSaver Scheme along with a future calculator and showing predictions for government contribution. This project was implemented to retire an old KiwiSaver application and Investment fund application to upgrade tech stack from Angular to react.

Responsibilities

- Requirement analysis, plan, design and development of customer facing web application using React, Redux and Typescript.

- Worked closely with product owners, UX designers and Business analysts to deliver secure and scalable solution aligned with banking compliance and regulatory standards.
- Worked on integrating and wiring applications to C# .NET Core REST APIs.
- Contributed to design system adoption and consistent UI patterns across platforms.
- Participated in release planning, deployments, and production support.
- Worked on developing and maintaining Rest API in C# .NET

Technology

- React, Redux, Redux saga, Storybook, Design system
- C# .NET Core

Role - Senior Software Engineer

[KiwiSaver Application](#), [Investment Fund Application](#),
[KiwiSaver Change Fund](#) - Financial investment Platform

[Auckland - NZ](#) Dec 2018 - May 2021

Project: Digital investment customer facing web platform for retirement or saving for your first home, platform helps customers to manage, grow and maximize savings. This project involved enhancing the product to provide new options for customers and fix the existing issues.

Responsibilities

- Requirement analysis, plan, design and development of customer facing web application using Angular, redux
- Worked closely with product owners, UX designers, and Business analysts to deliver new features and improve existing journeys.
- Designed and developed Rest API in C# .NET supporting investment transactions, fund switches, withdrawals, and account management
- Integrated 3rd party wealth calculators into application
- Participated in release planning, deployments, and production support

Technology

- Angular 2, React-Redux, C# .NET Core

[Centelon Limited](#)

[Nov 2017 – Nov 2018](#)

Company information – Centelon, a modern business-technology company, was born in this dynamic world of digital, cloud & analytics and carries a passion to help clients innovate, adapt and succeed.

Role - Senior Full Stack Developer

[MyDhan](#) – Mutual fund investment platform

[Pune - India, Auckland - NZ](#) March 2018 - Nov 2018

Project: MyDhan is a scalable, white-label capable, mutual fund investment platform enabling customer onboarding, portfolio management, transactions, and report generation along with processing Registrar and Transfer Agents (RTAs) files in context of BSE (Bombay Stock Exchange).

Responsibilities

- Led end-to-end delivery of the project, owning design, architecture, development, and deployment along with production support.
- Led and mentored team of 3 developers, providing technical guidance, code reviews, and delivery oversight.
- Functional specification and technical specification documentation with customer and admin portal capability.
- Designing frontend and backend architecture using MERN (MongoDB, Express.js, React and Node.js) with responsive, data-driven UI for scalable app development.
- Designed and implemented AWS services integration like S3, SNS (notification), SES (email), EC2.
- Integration with BSE (Bombay Stock Exchange) API platform and Aadhar based KYC integration.
- Implemented capability for uploading/downloading and processing RTA files in nightly job.
- Database design and architecture

Technology

- React, React-Redux, Typescript, Node.js, Express.js, Semantic-UI React, Bootstrap, HTML5-CSS

- MongoDB 3.4, Mongoose, AWS cloud (EC2, S3, SNS, SES)

Role – Senior Developer

Action Planner – Task Management Application

Pune, India Nov 2017-Mar 2018

Project: Task management platform enabling organizations to manage projects, assign tasks and track execution.

Responsibilities

- Assisted as a Senior full-stack developer to project development and enhancements.
- Built frontend using AngularJS and backend using node.js and express.js.
- Implemented authentication using AWS Cognito service and deployed to AWS using serverless.

Technology

- MEAN stack (AngularJS, Node.js, Express.js, MongoDB), Happy-js, Bootstrap, HTML5-CSS
- AWS (Serverless, Cognito)

Role – Senior Developer

ChurchConnect - Volunteer management for Church

Pune, India Nov 2017-Feb 2018

Project: White-labeled volunteer and community engagement platform for church's and non-profit organizations.

Responsibilities

- Assisted as senior developer solution designing, database design and rules-engine implementation.
- Implemented backend API using Node.js, Express.js and MongoDB.
- Mentoring new team members on MEAN stack
- Built Admin panel and server with session management and role-based access control.

Technology

- MEAN stack (AngularJS 1.6, NodeJS 4.0, Express, MongoDB 3.4)

FIS Global Solution (Sungard Solutions)

Jul 2012 - Oct 2017

Company information – FIS Global Solution (Sungard) is the world's largest global provider dedicated to financial technology solutions headquartered in Jacksonville, Florida. FIS empowers the financial world with software, services, consulting and outsourcing solutions focused on retail and institutional banking, payments, asset and wealth management, risk and compliance, trade enablement, transaction processing and record-keeping.

Role – Senior Engineer-Product Development

Securities360 – One platform for all product integration

Pune, India

Dec 2015-Oct 2017

Project: Securities360 is an integrated product with a rich and responsive UI developed in HTML5, AngularJS, Java Spring MVC and using the latest technology. Security360 product will allow access to multiple products at single place with single sign on (SSO) feature. We are migrating the Applet application into HTML5, Angular 1.x and Java Spring MVC. All functionality we are migrating to newer technologies and removing the dependencies on legacy libraries which was tightly coupled with the Applet product.

Responsibilities

- Development and enhancement of the Securities360 product from scratch.
- Requirement analysis, plan, design, implementation and unit testing.
- Involved in environment set up, deployment and maintenance.
- Key role Integrating legacy applet application to current application to support old functionality
- SSO integration
- Architecture and design flow of the product.
- Development of tools for L1 or L2 support.
- Integration of legacy applications into Securities360 to access them.
- Technical support for client testing.

Technology

- HTML5/CSS3, Bootstrap, AngularJS /2.x, NodeJS
- Java Spring MVC, REST Web Service

- MS SQL 2008 R2
- Microsoft VS 2013, Eclipse, Sublime Text, Visual Studio code, Node.js, Git

Role – Engineer-Product Development

Oasys - Trade Processing Platform

Pune, India Jul 2012-Mar 2017

Project: Oasys application supports manual and automated processing of trades in the Omgeo queue and can see the status of any trade. Also, platform allows user to add or update or remove the record from the processing queue manually and allows user to enable the flags for automatic processing.

Responsibilities

- Development and maintenance of Java Spring application.
- Requirement analysis, design, implementation and unit testing.
- Involved in enhancing new features into application and fixing the bugs.
- Functionality enhancements at front end.
- Involved in pre-installation requirements set up, deployment and maintenance.
- Live technical support for front end issues.

Technology

- Java, Spring Framework, HTML4, JavaScript, jQuery, Ajax, CSS3, REST Web service
- Eclipse, JAX-CXF, MS SQL 2008 R2

Role – Engineer-Product Development

FIX (Financial Information Exchange) – Trading & Settlement Platform **Pune, India** Jul 2012-Mar 2017

Project: Financial Information Exchange (FIX) is a universally accepted electronic communication standard protocol to exchange the financial information for trading, clearance and settlement. FIX is a console application that processes high volume data with high speed to the real time market with low latency.

Responsibilities

- Development, maintenance, and enhancement of the console application.
- Involved in requirement analysis, plan, design, implementation, and unit testing of new functionality.
- Worked on Installation, deployment, and maintenance.
- Key role in data, application, and server migration.
- Development of tools for support.
- Live technical support on production to resolve client issues.

Technology

- C++, FIX (Financial Information Exchange) protocol, MS SQL 2008 R2
- Microsoft VC++ 6.0, MSMQ, SVN

Role – Engineer-Product Development

Mutual Fund Market Tool (MFMT)

Pune, India Jan 2013-Apr 2014

Project: Mutual Fund Market Tool (MFMT) platform for trade data analysis

Mutual Fund Market Tool (MFMT) is a rich client web application designed using MVVM framework which has front end as Silverlight XAML and backend is REST web services. Application provides the consolidated view of the data from multiple entities. The application provides a UI which has all handy information, comparisons and related data gathered on single click.

Responsibilities

- Development, maintenance and enhancement of the console application.
- Development and maintenance of web applications.
- Requirement analysis, design and development of the application in .Net Silverlight MVVM framework.
- Interaction the business analyst for client requirements and functionality.
- Involved in deployment and maintenance.
- Involved in technical Support.
- Enhance Rest API developed in C# .NET

Technology

- C++, FIX (Financial Information Exchange) protocol, MS SQL 2008 R2
- Microsoft VC++ 6.0, MSMQ, SVN

- C# .NET, Silverlight

Awards & Recognitions

- **Top 5 Rank in FIS Hackathon 2017:** Selected in the top 5 for innovative idea implementation during the FIS Hackathon 2017.
- **Runner-Up in FIS Innov8 2017:** Achieved runner-up position for innovative idea implementation at the FIS Innov8 2017 event.

Education

- BE Information Technology, Pune University (India) in 2011 – First class – 60.26%

Certification & Training

- OCPJ Certification - *Issued by Oracle, 2012*
- Pluralsight trainings (React, Node.js, Redux, Angular.js, MongoDB, JavaScript)

Personal details

- Available for contract roles based in New Zealand or remote.
- NZ permanent resident visa.